Communication-Efficient Multi-Server Federated Learning via Over-the-Air Computation

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***Abstract*—Thanks to the Internet of Things (IoT), there has been explosive growth in edge devices, which generate a tremendous amount of data that holds invaluable potential. However, conventional data mining and machine learning (ML) paradigms require transmitting raw data to data centers for further use, which puts a heavy burden on communication networks and is exposed to high privacy risks. Federated learning allows for the training of ML models using distributed datasets, which can be applied to protect data privacy and alleviate transmission burdens. Meanwhile, the technique of over-the-air (OTA) computation can be utilized to exploit the superposition property of wireless communication channels. Motivated by this, in this paper, we propose a co-phase OTA approach for communication-efficient uploading in multi-server federated learning, which does not require expansion of the uplink channel bandwidth when the numbers of users and models increase. Besides, the digital OTA with randomized transmission is proposed to overcome the disadvantages of analog OTA, where the performance analyses of analog OTA and digital OTA are deduced, respectively. Simulation results show that a lower cost function can be obtained by digital OTA while requiring fewer iterations for convergence than that in analog OTA as more users can upload.**

***Index Terms*—Federated learning, over-the-air (OTA) computation, distributed stochastic gradient descent, Internet-of-Things.**

I. Introduction

RECENTLY, the Internet-of-Things (IoT) plays a crucial role in our daily life. With the large-scale deployment of edge IoT devices, edge data have ushered in explosive growth, including images and videos, which has inestimable potential value [1]. However, conventional data mining and machine learning paradigms require transmitting raw data to data centers for further use, which puts a heavy burden on communication networks and is exposed to high privacy risks. Federated learning [2], [3], [4] enables the training of machine learning models using distributed datasets that reside on mobile edge devices, where exists a server and multiple users with datasets in federated learning. Since mobile edge users in federated learning do not need to upload their datasets, thereby protecting the data privacy. However, the number of devices is typically large in IoT systems, leading to a high-scale optimization challenge in federated learning, which poses difficulties in distributed optimization of large-scale learning models [5], [6]. Researchers have proposed several approaches to address these problems, e,g., quantization of gradients, sparsification of gradients, and skipping communication rounds [7], [8]. Besides, the authors in [9] have proposed a Qsparse-local-Stochastic Gradient Descent (SGD) algorithm for modern large-scale optimization as applied to machine learning, which combines aggressive sparsification with quantization and local computations. Meanwhile, in [10], hierarchical federated learning achieves a high communication efficiency between cloud servers and edge servers by leveraging a client-edge-cloud aggregation hierarchy.

In federated learning, the server needs to train a machine learning model using datasets from multiple users through iterative processes. During each iteration, the server disseminates the current parameter vector to users, then users compute their local gradient vectors using their datasets and transmit them back to the server. Subsequently, the server computes the average of the local gradient vectors and updates the parameter vector for the next iteration. Therefore, both the number of users and the number of iterations contribute to the increase in communication cost. Some works have been studied to reduce the number of iterations, for example, in [11], the authors proposed a federated averaging (FedAvg) method, where each user can update the parameter vector locally using their own dataset. In [12], a second-order over-the-air federated optimization algorithm was proposed, which can reduce the communication rounds of global model aggregation with low-latency. Meanwhile, some researchers reduced the communication overhead by limiting the number of users in a single access [13] and some others chose using non-orthogonal multiple access (NOMA) in federated learning to enhance communication efficiency [14], [15], [16].

Meanwhile, the technique of over-the-air (OTA) computation has been proposed to utilize the superposition property of wireless communication channels for mobile edge users [17], [18]. In OTA, it enables simultaneous transmissions of users' local updates through a shared channel, leading to the aggregation at the server. There are a number of studies that adopted OTA approaches to wireless federated learning, some of them require

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the channel state information (CSI) [19], [20], [21], others do not require CSI [22]. In IoT applications, federated learning can also be effectively utilized with a vast number of devices, employing a random access protocol for uplink transmissions [23], [24], [25].

Moreover, applying federated learning on edge devices can help create an edge artificial intelligence (AI) system [26], which is able to deal with the challenges of trustworthiness and scalability in the 6th generation (6G) [27], [28]. In addition, some of the OTA approaches attempt to transmit coding signals with a linear structure, which helps receivers recovering the transmitted codewords more reliable [29], [30]. In order to achieve energy-efficient and power-control policies, device selection strategies and dynamic device scheduling mechanisms are utilized in [31], [32]. Furthermore, a cooperative power control scheme is presented in [33] to minimize the weighted sum mean squared error (MSE) in a multi-cell OTA computation system.

In order to meet the requirement of a straightforward generalization of federated learning, the bandwidth of uplink channel for users' uploading should expand by a factor of the number of servers. Thus, the approach in [34][1] is a generalization of the approach in [23], where multi-channel random access can be used and multiple channels are assigned to multiple models. Meanwhile, in [35], we jointly used co-phase (CP) transmission and OTA aggregation to reduce bandwidth requirements. In this paper, we extend our previous work in [35] where the communication-efficient approach is based on the CP transmission and requires no bandwidth expansion of uplink channel. Moreover, we analyze the method by considering two new directions. Specifically, we analyze the convergence of the proposed methods and the corresponding error upper bound in detail and further analyze the impact of system parameters on the performance of our proposed method. Through the exploration of the number of users and servers, and the iteration step size, the superiority of the proposed method is further verified. The main contributions of the paper are three-fold as follows.

i) We present a CP-OTA approach for achieving communication-efficient uploading in multi-server federated learning. Our approach eliminates the need for bandwidth expansion of the uplink channel;
ii) We derive two key conditions that ensure an unbiased estimate of the aggregation at each server. Furthermore, we propose power control and model selection methods to enhance the performance of the system;
iii) A digital OTA approach based on randomized transmission is proposed, where the performance analysis of analog OTA and digital OTA is carried out. Simulation results show that a lower cost function can be obtained in digital OTA while requiring fewer iterations for convergence than that in analog OTA as more users can upload.

The rest of the paper is organized as follows. We present the system model with multiple servers in Section II. In Section III, we generalize the analog OTA approach, where the power control and model selection are proposed. After identifying two main disadvantages of analog OTA, we then propose a digital OTA approach based on randomized transmission in Section IV. In Section V, we derive the performance analysis with/without randomized transmission. We show the simulation results in Section VI and conclude the paper with some remarks in Section VII. For convenience, the list of key symbols is given in Table I.

Notation: Matrices and vectors are denoted by upper- and lower-case boldface letters, respectively. $\mathbb{E}[\cdot]$ and $\mathrm{Var}(\cdot)$ denote the statistical expectation and variance, respectively. $\mathcal{CN}(a, R)$ represents the distribution of circularly symmetric complex Gaussian (CSCG) random with mean a and covariance R.

[1] In this work, an access point (AP) or base station has multiple models, which differs from the scenario in this paper where each AP/server is dedicated to its own model.

TABLE I
SUMMARY OF MAIN KEY SYMBOLS

Notation	Definition
N_U	The number of users
N_S	The number of servers/models/APs
L	The length of parameter vectors
C_s	The cost function of model s
$\mathcal{D}$	The training dataset of the system
$\mathbf{g}_{u,s}^{(t)}$	The gradient vector of the cost of model s at user u at the tth iteration
$\mathbf{r}_s(t)$	The received signal at AP s in slot t
$\mathbf{x}_u(t)$	The signal sequence transmitted by user u

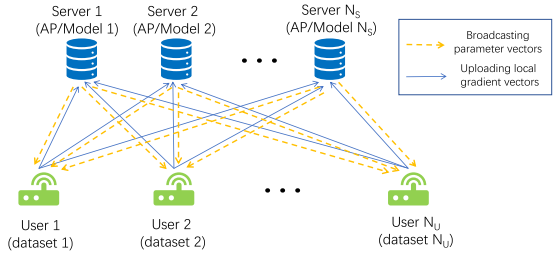


Fig. 1. Multi-model federated learning with N_S APs.

II. System Model

In this section, we consider a simple generalization of wireless federated learning with multiple servers, where each server has its own model and is associated with own AP, as shown in Fig. 1. In this figure, solid lines represent downlink transmissions from servers to users and dashed lines represent uplink transmissions from users to servers. Each user and server own independent transceiver antennas. Meanwhile, each user has its own dataset and each server trains its own model using shared distributed datasets residing on mobile users [34]. For convenience, in this paper we assume that each server has one model, which means that models, servers, and APs are all interchangeable.

Assume that there are N_S servers that are associated with their own APs. For training, the cost function can be represented by C_s for server s. Let N_U denote the number of users and $\mathcal{D}_u$ the dataset at user u. In addition, let $|\mathcal{D}_u|$ denote the size of each user's dataset, which is the same for convenience. Then,

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the optimal parameter vector for model s is

$$\mathbf{w}_s^* = \underset{\mathbf{w}_s}{\operatorname{argmin}} \frac{1}{N_U} \sum_{u=1}^{N_U} C_s(\mathbf{w}_s; \mathcal{D}_u),\ s = 1, \ldots, N_S. \quad (1)$$

Here, we assume that $\mathbf{w}_s \in \mathbb{R}^L$ for all s, where L represents the length of parameter vectors. Throughout the paper, we assume that $N_U \gg N_S$, i.e., the number of users is much larger than the number of servers.

In federated learning, the process of transmission operates in the time division duplex (TDD) mode, where the time is slotted and each time slot is divided into two sub-slots. The first sub-slot is dedicated to transmissions from APs to users, e.g., the sth AP transmits its orthogonal pilot signals as well as the updated parameter vector, denoted by $\mathbf{w}_s^{(t)}$, within time slot t. While the second sub-slot is reserved for transmissions from users to APs, at user u, the following gradient vector can be obtained for model s:

$$\mathbf{g}_{u;s}^{(t)} = \nabla \mathbf{C}_s \left(\mathbf{w}_s^{(t)}, \mathcal{D}_u \right), \quad (2)$$

where $\nabla f(\mathbf{x})$ represents the gradient vector of function $f(\mathbf{x})$. It is worth noting that the norm of the local gradient vector should be finite as it needs to be uploaded in federated learning, i.e.,

$$||\mathbf{g}_{u;s}^{(t)}||^2 \leq \Delta_s, \quad (3)$$

where $\Delta_s < \infty$. To this end, the cost function C_s is assumed to be β_s-smooth [36]. Letting $\mathbf{g}_s = \frac{1}{N_U} \sum_{u=1}^{N_U} \mathbf{g}_{u;s}$, it can be shown that

$$||\nabla \mathbf{C}_s(\mathbf{w}, \mathcal{D}) - \nabla \mathbf{C}_s(\mathbf{w}_s^*, \mathcal{D})|| \leq \beta_s ||\mathbf{w} - \mathbf{w}_s^*||,$$

and since $\nabla \mathbf{C}_s(\mathbf{w}_s^*, \mathcal{D}) = \mathbf{0}$ and $\nabla \mathbf{C}_s(\mathbf{w}_s, \mathcal{D}) = \mathbf{g}_s$, we can get

$$||\nabla \mathbf{C}_s(\mathbf{w}, \mathcal{D})|| \leq \beta_s ||\mathbf{w} - \mathbf{w}_s^*||,$$

Therefore, as long as $\mathbf{w}_s^{(t)}$ and $\mathbf{w}^*$ are finite, (3) holds.

In the second sub-slot, we assume that AP s receives the gradient vectors from users without errors, and the parameter vector for model s can be updated as:

$$\mathbf{w}_s^{(t+1)} = \mathbf{w}_s^{(t)} - \mu \mathbf{g}_s^{(t)}, \quad (4)$$

where $\mathbf{g}_s^{(t)} = \frac{1}{N_U} \sum_{u=1}^{N_U} \mathbf{g}_{u;s}^{(t)}$ represents the aggregation in time slot t and μ denotes the step size or learning rate.

As discussed above, to update the parameter vectors as in (4), there should be $N_U N_S$ uplink transmissions per iteration. In practice, to avoid interference between the channels between different users and the server, the required uplink bandwidth is proportional to the number of users, N_U and that of servers, N_S. However, the uplink bandwidth is limited, which will hinder increasing the number of users to obtain a better training result. To solve this problem, various OTA approaches are considered in [19], [20], [21], [22].

III. CP Aggregation

In this section, we present a generalized approach for analog OTA that eliminates the need for bandwidth extension. In addition, we also propose the power control and model selection algorithm in the system.

A. Analog OTA Aggregation

Thanks to TDD, the OTA aggregation can take advantage of channel reciprocity using known (or estimated) CSI. For stationary users and servers, large-scale fading is constant across multiple time slots, while small-scale fading coefficients can vary between slots. Let $h_{u;s}(t)$ denote the channel coefficient between user u and AP s in time slot t, and assume that it is symmetric Rayleigh fading [37], which can be given by

$$h_{u;s}(t) \sim \mathcal{CN}(0, \sigma_h^2), \quad (5)$$

where σ_h^2 denotes the variance of the channel fading. User u can estimate $h_{u;s}(t)$, $s = 1, \ldots, N_S$, when the APs transmit their pilots during the first sub-slot. Then, the received signal at AP s during the second sub-slot is

$$\mathbf{r}_s(t) = \sum_{u=1}^{N_U} h_{u;s}(t) \mathbf{x}_{u,s}(t) + \mathbf{n}_s(t), \quad (6)$$

where $\mathbf{x}_{u,s}(t) \in \mathbb{R}^L$ denotes the signal sequence transmitted by user u to AP s and $\mathbf{n}_s(t) \sim \mathcal{CN}(0, N_0 \mathbf{I})$, where $\mathbf{n}_s(t) \in \mathbb{C}^L$ is the background noise at AP s.

In the following, we generalize the analog OTA approach, which does not necessitate bandwidth extension as the number of users increases. Suppose that the uplink channel is divided into N_S orthogonal channels, where each AP is allocated a dedicated uplink channel for uploading data using OTA transmission. Then, user u transmits the local gradient vector of model s through channel s, which can be shown as:

$$\mathbf{x}_{u;s}(t) = \sqrt{P_{u;s}(t)} \alpha_{u;s}(t) \mathbf{g}_{u;s}^{(t)}, \quad (7)$$

where $\alpha_{u,s}(t) = \frac{h_{u;s}^*(t)}{|h_{u;s}(t)|}$ ($h_{u;s}^*(t)$ denotes the conjugate matrix of the channel) and $P_{u;s}(t)$ denotes the transmit power of user u in channel s at time slot t. At AP s, the received signal through (orthogonal) channel s becomes

$$\mathbf{r}_s(t) = \sum_{u=1}^{N_U} \sqrt{P_{u;s}(t)} |h_{u;s}(t)| \mathbf{g}_{u;s}^{(t)} + \mathbf{n}_s(t). \quad (8)$$

B. CP Transmission

As indicated in (8), while the uplink bandwidth does not scale with the number of users, it does increase with the number of models, N_S. In this subsection, we discuss CP transmission, which allows for the utilization of a single uplink bandwidth regardless of N_S.

Since Rayleigh fading is assumed as in (5), when the fading coefficients' phases are independent and uniformly distributed, the correlation of channel coefficients becomes zero, i.e.,

$$\mathbb{E}\left[h_{u;s}(t) \alpha_{u;s'}(t)\right] = 0,\ s \neq s', \quad (9)$$

which can be used to reduce N_S orthogonal channels to a shared single channel. To accomplish this, let us assume that each user transmits a superposition of N_S local gradient vectors through a shared uplink channel. This approach is known as the CP aggregation. In this case, the signal to be transmitted by user

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u with power P_u can be expressed as follows:

$$\mathbf{x}_u(t) = \sqrt{P_u} \sum_{s=1}^{N_S} \alpha_{u;s}(t) \mathbf{g}_{u;s}^{(t)}, \tag{10}$$

and the received signal at AP s is

$$\mathbf{r}_s(t) = \sum_{u=1}^{N_U} \sqrt{P_u} |h_{u;s}(t)| \mathbf{g}_{u;s}^{(t)} + \sum_{u=1}^{N_U} \sqrt{P_u} \sum_{s' \neq s} h_{u;s}(t) \alpha_{u;s'} \mathbf{g}_{u;s'}^{(t)} + \mathbf{n}_s(t). \tag{11}$$

For convenience, we assume that $\mathbf{g}_{u;s}^{(t)}$ is real-valued and $\mathbf{r}_s(t)$ and $\mathbf{n}_s(t)$ represent the real-valued terms only. Thus, we only need to take the real terms of $\mathbf{r}_s(t)$ in (11), Therefore, $\mathbf{n}_s(t) \sim \mathcal{N}(0, \frac{N_0}{2}\mathbf{I}_L)$ and $\mathbf{n}_s(t) \in \mathbb{R}^L$.

Then, each user performs model selection, where it chooses one model among N_S models to upload the local gradient vector. The signal sequence $\mathbf{x}_u(t)$ becomes

$$\mathbf{x}_u(t) = \sqrt{P_u} \alpha_{u;s(u)} \mathbf{g}_{u;s(u)}^{(t)}, \tag{12}$$

where $s(u) \in \{1, \ldots, N_S\}$ stands for the index of the selected model by user u.

Let $\mathcal{K}_s$ represent the set of the users that choose model s, i.e., $\mathcal{K}_s = \{u : \ s(u) = s, \ u = 1, \ldots, N_U\}$. For convenience, we will omit the time index t. Then, the received signal at AP s becomes

$$\mathbf{r}_s = \sum_{u \in \mathcal{K}_s} \sqrt{P_u} |h_{u;s}| \mathbf{g}_{u;s} + \sum_{u \in \mathcal{K}_s^c} \sqrt{P_u} \sum_{s' \neq s} h_{u;s} \alpha_{u;s'} \mathbf{g}_{u;s'} + \mathbf{n}_s, \tag{13}$$

where $\mathcal{K}_s^c$ is the complement set of $\mathcal{K}_s$. In (13), P_u of user u choosing model s can be given by

$$\sqrt{P_u} = \frac{c}{|h_{u;s}|}. \tag{14}$$

Here, $c > 0$ is a constant to be decided for the strict power constraint $P_u \leq P_{\max}$, where $P_{\max}$ denotes the maximum transmitting power of the edge users. Then, if each user u chooses a model uniformly at random, for a given model s, the probability that user u chooses model s is $\frac{1}{N_S}$. Thus, the mean of $\mathbf{r}_s$ becomes

$$\mathbb{E}[\mathbf{r}_s] = \mathbb{E}[\mathbb{E}[\mathbf{r}_s \,|\, \mathcal{K}_s]] = c \frac{1}{N_S} \sum_{u=1}^{N_U} \mathbf{g}_{u;s}. \tag{15}$$

Then, we can know that

$$\mathbb{E}[\mathbf{r}_s] \propto \mathbf{g}_s.$$

Note that the size of the mini-batch, denoted as $|\mathcal{K}_s|$, can be considered. If $|\mathcal{K}_s|$ is selected randomly in each iteration, the resulting update process, which involves random model selection, can be identified as the SGD method [38], which has the following updating rule (instead of (4)) at AP s:

$$\mathbf{w}_s^{(t+1)} = \mathbf{w}_s^{(t)} - \mu \mathbf{r}_s(t). \tag{16}$$

Consequently, in each iteration, the following steps are carried out for CP aggregation with OTA:

1) (Downlink) AP s sends the parameter vector $\mathbf{w}_s^{(t)}$ and its pilot signal. Since there are N_S APs/severs, it is assumed that the first sub-slot is further divided into N_S mini-slot and the sth mini-slot is allocated for the sth AP.
2) (Uplink) Each user computes its gradient vectors for all models, chooses a model, and performs CP OTA by sending $\mathbf{x}_u(t)$ in (12) through a shared single uplink channel.

In summary, using CP aggregation with OTA, we can show that a significant uplink bandwidth reduction can be achieved as follows:

$$N_U N_S \underset{\text{Over-the-Air}}{\rightarrow} N_S \underset{\text{CP/Model selection}}{\rightarrow} 1 \tag{17}$$

C. Power Control and Model Selection

Considering that the transmit power of users may be limited, the system adopts truncated channel inversion, which is an adaptive power control strategy. For truncated channel inversion, the transmitter inverts the channel gain to maintain a constant received power only when the channel quality is above a threshold, and suspends transmission otherwise to avoid excessive power consumption. By setting a channel gain threshold, the system performs channel inversion selectively to balance performance and energy efficiency, and optimize capacity and power utilization. The truncated channel inversion can be given by

$$\sqrt{P_u} = \begin{cases} \frac{c}{|h_{u;s}(t)|}, & \text{if } \frac{c^2}{|h_{u;s}(t)|^2} \leq P_{\max} \\ 0, & \text{otherwise.} \end{cases} \tag{18}$$

However, this truncated channel inversion policy cannot be directly applied. Since the actual transmit power is $(\sqrt{P_u}[\mathbf{g}_{u;s}]_l)^2$, $l = 1, \ldots, L$, where $[\mathbf{x}]_l$ denotes the lth element of vector $\mathbf{x}$, the practical truncated channel inversion should be

$$\sqrt{P_u} = \begin{cases} \frac{c}{|h_{u;s}(t)|}, & \text{if } \frac{\max_l (c[\mathbf{g}_{u;s}]_l)^2}{|h_{u;s}(t)|^2} \leq P_{\max} \\ 0, & \text{otherwise,} \end{cases} \tag{19}$$

Note that the (peak) signal-to-noise ratio (SNR) at AP s from user u without the interference is the sum of the other signals transmitted to the other APs (or $N_S = 1$ is assumed), which can be given by

$$\text{SNR}_s = \frac{P_u |h_{u;s}|^2 \max_l |[\mathbf{g}_{u,s}]_l|^2}{N_0/2} = \frac{c^2 \max_l |[\mathbf{g}_{u,s}]_l|^2}{N_0/2} \leq \frac{c^2 \Delta_s}{N_0/2}. \tag{20}$$

Due to the presence of the background noise, $\mathbf{n}_m$, the resulting SGD in (16) becomes the perturbed SGD [39], [40], and the SNR in (20) can be decided by c to adjust the influence of the perturbation. Note that if $N_S > 1$, together with the background noise, the interference can serve as perturbation.

In order to make $\mathbf{r}_s$ an unbiased estimate of $\mathbf{g}_s$ through CP aggregation and SGD, the condition, e.g., each user u chooses a model uniformly at random, should be satisfied. In this case, from (5) and (19), the outage probability can be represented by the probability that user u does not transmit its local gradient

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vector, which is given by

$$\begin{aligned}\mathbb{P}_{\text{out}} &= \Pr\left(|h_{u;s}(t)|^2 < \frac{\max_l (c[\mathbf{g}_{u;s}]_l)^2}{P_{\max}}\right)\\ &= 1-\exp\left(-\frac{\max_l (c[\mathbf{g}_{u;s}]_l)^2}{\sigma_h^2 P_{\max}}\right)\\ &\le 1-\exp\left(-\frac{c^2||\mathbf{g}_{u;s}||^2}{\sigma_h^2 P_{\max}}\right)\\ &\le 1-\exp\left(-\frac{c^2\Delta_s}{\sigma_h^2 P_{\max}}\right), \end{aligned} \tag{21}$$

where the first inequality is due to $\max_l |[\mathbf{x}]_l|^2 \le ||\mathbf{x}||^2$ and the second inequality is due to $||\mathbf{g}_{u;s}||^2 \le \Delta_s$. From (20), it can be shown that

$$\kappa = \frac{c^2\Delta_s}{\sigma_h^2 P_{\max}} \ge \frac{\text{SNR}_s}{\Gamma_{\max}}, \tag{22}$$

where $\Gamma_{\max} = \frac{\sigma_h^2 P_{\max}}{N_0/2}$ is the average received SNR when a user transmits the signal at the maximum transmit power (here, the average takes over the Rayleigh fading channel in (5)). From (21), in order to have a sufficiently low outage probability, a small κ is expected. Since $\Gamma_{\max}$ is given by the channel and communication system (i.e., $P_{\max}$), a certain value of κ can be obtained by adjusting c.

As proposed in [19], it is suggested to increase the number of users transmitting local gradient vectors. To minimize the outage probability $\mathbb{P}_{\text{out}}$, user u can adopt for the following model selection:

$$s(u) = \underset{s}{\text{argmin}} \frac{\max_l ([\mathbf{g}_{u;s}]_l)^2}{|h_{u;s}(t)|^2}. \tag{23}$$

Assuming the statistical properties of $\mathbf{g}_{u;s}$ are the same for all s, the probability of selecting model s becomes $\frac{1}{N_S}$ due to the symmetric condition. Moreover, when $\Delta_s = \Delta$ for all s (or letting $\Delta = \max_s \Delta_s$), the outage probability can be demonstrated as follows:

$$\begin{aligned}\mathbb{P}_{\text{out}} &= \Pr\left(\min_s \frac{c^2 \max_l ([\mathbf{g}_{u;s}]_l)^2}{|h_{u;s}(t)|^2} > P_{\max}\right)\\ &\le \Pr\left(\min_s \frac{c^2\Delta_s}{|h_{u;s}(t)|^2} > P_{\max}\right)\\ &= \Pr\left(\max_s |h_{u;s}(t)|^2 < \frac{c^2\Delta}{P_{\max}}\right)\\ &\le \left(1-e^{-\kappa}\right)^{N_S}. \end{aligned} \tag{24}$$

Hence, the upper-bound of outage probability stated in (24) has the potential to decrease as N_S increases, thanks to the optimal model selection specified in (23). Here, N_S represents the diversity gain achieved through model selection. For simplicity, we refer to the resulting method as analog co-phase OTA with model selection (AOTA-MS).

D. Disadvantages of Analog OTA

Although we have reduced the uplink channel requirements of the federated learning system by using the CP OTA technology, there are still some shortcomings of the CP OTA technology.

The first one is the user needs a radio frequency (RF) linear power amplifier to transmit a local gradient vector, where the transmit power is $(\sqrt{P_u}[\mathbf{g}_{u;s}]_l)^2 = \frac{(c[\mathbf{g}_{u;s}]_l)^2}{|h_{u;s}|^2}, l = 1, \ldots, L$. Thus, the cost of the linear power amplifier could be high when the transmit power becomes large. The second one involves the utilization of a truncated channel inversion policy. However, this approach may exhibit an undesirable characteristic where the initial stages of training have a small number of users transmitting local gradient vectors, while this number gradually increases as convergence occurs. Consequently, it results in a slow convergence speed.

In addition to the above main disadvantages, there is another disadvantage in deciding the value of c to guarantee a certain outage probability. As shown in (21) and (24), for a given $\Gamma_{\max}$, it is necessary to know Δ_s in advance so that the value of c can be determined to meet a target outage probability. However, since Δ_s is unfortunately unknown in advance, it may not be possible to guarantee a target outage probability.

IV. Randomized Transmission for Soft-Gain in Digital OTA

In this section, we introduce randomized transmission that can mitigate the problems discussed in Section III-D.

A. Advantage of Digital OTA

Although the implementation of analog OTA computing is relatively simple, it cannot effectively combat noise. Especially when the transmit power is low, the signal received by the server can have large interference. Therefore, to avoid those disadvantages stated in Section III-D, a digital CP-OTA approach should be fully studied. Beyond the advantages of the analog OTA mentioned in [19], the digital CP-OTA approach does not require high-fidelity linear power amplifiers at the user to mitigate noise and interference.

Regarding digital OTA systems, there are one-bit quantization as mentioned in [21], and randomized quantization as discussed in [7] and [20]. In particular, randomized quantization can be employed as it provides an unbiased estimation of the local gradient vector. However, it necessitates the design of a codebook. Meanwhile, a randomized transmission approach can be employed without codebook design by using the following transmission probability, which is

$$p_{u;s,l} = \min\left\{1, \gamma|[\mathbf{g}_{u;s}]_l|\right\}, \tag{25}$$

where $\gamma > 0$ denotes a control parameter, which is the probability that the lth element of $\mathbf{g}_{u;s}$ is transmitted.

Then, the lth element of the signal sequence to be transmitted by user u becomes

$$z_{u;s,l} = \text{sign}\left([\mathbf{g}_{u;s}]_l\right) \mathbb{1}\!\!\nVdash (U_{u;s,l} \le p_{u;s,l}) \in \{-1, 0, +1\}, \tag{26}$$

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where $U_{u;s,l} \sim U(0,1)$ is a uniform random variable, $\mathbb{1}$ denotes the indicator function, and $\text{sign}(\cdot)$ means to take the same sign. Let $\mathbf{z}_{u;s} = [z_{u;s,1} \ \dots \ z_{u;s,L}]^{\mathrm{T}}$, if γ is sufficiently small, it can be shown that

$$\mathbb{E}[\mathbf{z}_{u;s}] = \gamma \mathbf{g}_{u;s}. \tag{27}$$

The method used to generate $\mathbf{z}_{u;s}$ is referred to as randomized transmission for soft-gain, which can also be interpreted as a form of randomized quantization.

B. Randomized Transmission for Digital OTA

In the digital CP-OTA approach, the CSI can be known through the channel reciprocity. Let $h_{u;s}(t) \sim \mathcal{CN}(0, \sigma_h^2)$ denote the symmetric Rayleigh fading channel coefficient between user u and AP s in time slot t, the received signal at AP s during the second sub-slot is

$$\hat{\mathbf{r}}_s(t) = \sum_{u=1}^{N_U} h_{u;s}(t)\hat{\mathbf{x}}_{u,s}(t) + \mathbf{n}_s(t), \tag{28}$$

where $\hat{\mathbf{x}}_{u,s}(t)$ denotes the signal sequence transmitted by user u to AP s and $\mathbf{n}_s(t) \sim \mathcal{CN}(0, N_0\mathbf{I})$ is the background noise at AP s. Here, $\hat{\mathbf{x}}_{u,s}(t) \in \mathbb{R}^L$and $\mathbf{n}_s(t) \in \mathbb{C}^L$.

Then, user u transmits the local gradient vector of model s through channel s via digital OTA system as follows:

$$\hat{\mathbf{x}}_{u;s}(t) = \sqrt{P_u}\alpha_{u;s}(t)\mathbf{z}_{u;s}^{(t)}, \tag{29}$$

where P_u represents the transmit power of user u. If we set all the transmit powers to be the same, with $z_{u;s,l} \in \{-1, 0, +1\}$, the transmit power becomes independent of $\mathbf{g}_{u;s}$. It means that the transmit power is solely decided by the CSI, and the power control policy in (18) can be used. With the randomized transmission for soft-gain, the system realizes randomized quantization without codebook and eliminates restrictions on the utilization of truncated channel inversion for analog OTA.

At AP s, the received signal through channel s becomes

$$\hat{\mathbf{r}}_s(t) = \sum_{u=1}^{N_U} \sqrt{P_u}|h_{u;s}(t)|\mathbf{z}_{u;s}^{(t)} + \mathbf{n}_s(t). \tag{30}$$

Thus, similar to (13)–(15), the mean of $\hat{\mathbf{r}}_s$ becomes

$$\begin{aligned}\mathbb{E}[\hat{\mathbf{r}}_s] &= \mathbb{E}\left[\mathbb{E}\left[\hat{\mathbf{r}}_s \,|\, \mathcal{K}_s\right]\right] \\ &= c\frac{\gamma}{N_S}\sum_{u=1}^{N_U} \mathbf{g}_{u;s} \propto \mathbf{g}_s.\end{aligned} \tag{31}$$

In addition, user u chooses model s as follows:

$$\begin{aligned}s(u) &= \underset{s}{\text{argmin}} \frac{1}{|h_{u;s}(t)|^2} \\ &= \underset{s}{\text{argmax}} |h_{u;s}(t)|^2,\end{aligned} \tag{32}$$

and the outage probability becomes:

$$\begin{aligned}\hat{\mathbb{P}}_{\text{out}} &= \Pr\left(\min_s \frac{c^2}{|h_{u;s}(t)|^2} > P_{\max}\right) \\ &= \Pr\left(\max_s |h_{u;s}(t)|^2 < \frac{c^2}{P_{\max}}\right) \\ &= \left(1 - \exp\left(-\frac{c^2}{\sigma_h^2 P_{\max}}\right)\right)^{N_S}.\end{aligned} \tag{33}$$

As discussed above, the average number of users transmitting their local vectors is the same for all models, which increases with the number of models, N_S. In addition, the outage probability is independent of Δ_s, while c can be adjusted to guarantee a certain target outage probability. The resulting approach is called digital CP-OTA with model selection and randomized transmission (DOTA-MSRT).

V. Performance Analysis

In this section, we analyze the performance of the method without randomized transmission and with randomized transmission, respectively.

A. Without Randomized Transmission

From (16), we have

$$\mathbf{w}_s^{(t+1)} = \mathbf{w}_s^{(t)} - \mu\mathbb{E}\left[\mathbf{r}_s(t)\right] - \mu\left(\mathbf{r}_s(t) - \mathbb{E}\left[\mathbf{r}_s(t)\right]\right). \tag{34}$$

Assuming that the noise term in (34), i.e., $\mu(\mathbf{r}_s(t) - \mathbb{E}[\mathbf{r}_s(t)])$, has a finite variance (we will discuss its variance later), it is desirable to find μ such that $||\mathbb{E}[\mathbf{w}_s^{(t+1)}] - \mathbf{w}_s^*|| \to 0$.

According to the power control and model selection discussed in Section III-C, here we use $\tilde{\mathcal{K}}_s$ to present the set of users choosing model s and transmitting its local gradients, its formula is given by

$$\begin{aligned}\tilde{\mathcal{K}}_s = \Bigg\{u : &\frac{\max_l([\mathbf{g}_{u;s}]_l)^2}{|h_{u;s}(t)|^2} \le \frac{\max_l([\mathbf{g}_{u;s'}]_l)^2}{|h_{u;s'}(t)|^2}, \ \forall s' \ne s, \\ &\text{and } \frac{c^2 \max_l([\mathbf{g}_{u;s}]_l)^2}{|h_{u;s}(t)|^2} \le P_{\max}\Bigg\},\end{aligned} \tag{35}$$

while $\tilde{\mathcal{K}}_s \subseteq \mathcal{K}_s$. Thus, $\mathbb{E}[\mathbf{r}_s]$ in (15) is modified as follows:

$$\begin{aligned}\mathbb{E}[\mathbf{r}_s] &= \mathbb{E}\left[\mathbb{E}\left[\mathbf{r}_s \,|\, \tilde{\mathcal{K}}_s\right]\right] \\ &= \frac{c\rho N_U}{N_S}\mathbf{g}_s,\end{aligned} \tag{36}$$

where $\rho = 1 - \mathbb{P}_{\text{out}} \le 1$. In addition to the assumption of β_s-smooth, we assume that C_s is θ_s-strongly convex, where $\theta_s < \beta_s$. Then, according to the gradient descent and the mean value theorem [36], let $\bar{\mathbf{w}}_s^{(t)} = \mathbb{E}[\mathbf{w}_s^{(t)}]$, we have

$$||\bar{\mathbf{w}}_s^{(t+1)} - \mathbf{w}_s^*|| = ||\bar{\mathbf{w}}_s^{(t)} - \mathbf{w}_s^*|| \cdot ||\mathbf{I} - \frac{\mu c\rho N_U}{N_S}\nabla^2 C_s||, \tag{37}$$

where $\nabla^2 C_s$ represents the Hessian at $\mathbf{w}_s'$ lies within the line segment that connects between $\bar{\mathbf{w}}_s^{(t)}$ and $\mathbf{w}_s^*$. Since $\theta_s\mathbf{I} \le \nabla^2 C_s \le \beta_s\mathbf{I}$, we can find the following condition for the convergence in mean:

$$\max\left\{1 - \frac{\mu c\rho N_U}{N_S}\theta_m, \frac{\mu c\rho N_U}{N_S}\beta_s - 1\right\} \le 1. \tag{38}$$

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Thus, with

$$\mu < \bar{\mu} = \frac{2N_S}{c\rho N_U(\theta_s + \beta_s)}, \tag{39}$$

since

$$||\mathbf{I} - \frac{\mu c\rho N_U}{N_S}\nabla^2 C_s|| < \frac{\beta_s - \theta_s}{\beta_s + \theta_s} \leq 1,$$

we can see that $\bar{\mathbf{w}}_s^{(t)} \to \mathbf{w}_s^*$ as $t \to \infty$ from (37).

In (34), omitting the time index t, the noise term can be divided into two terms as follows:

$$\begin{aligned}\mathbf{d}_s &= \mathbf{r}_s - \mathbb{E}[\mathbf{r}_s] \\ &= \mathbf{u}_s + \mathbf{e}_s,\end{aligned} \tag{40}$$

where $\mathbf{u}_s = \sum_{u\in\tilde{\mathcal{K}}_s} c\mathbf{g}_{u;s} - \mathbb{E}[\mathbf{r}_s]$ and

$$\mathbf{e}_s = \sum_{s'\neq s}\sum_{u\in\tilde{\mathcal{K}}_{s'}} \sqrt{P_u} h_{u;s}\alpha_{u;s'}\mathbf{g}_{u;s'} + \mathbf{n}_s. \tag{41}$$

Since $h_{u;s}$ and $\alpha_{u;s'}$ are independent, we have

$$\begin{aligned}\mathbb{E}\left[||\mathbf{e}_s||^2\right] &= \mathbb{E}\Big[\mathbb{E}\left[||\mathbf{e}_s||^2 \,|\, \tilde{\mathcal{K}}_1, \ldots, \tilde{\mathcal{K}}_{N_S}\right]\Big] \\ &= \sum_{s'\neq s}\mathbb{E}\left[\sum_{u\in\tilde{\mathcal{K}}_{s'}} \frac{P_u||\mathbf{g}_{u;s'}||^2|h_{u;s}|^2}{2}\right] + \frac{N_0 L}{2} \\ &\leq \sum_{s'\neq s}\mathbb{E}\left[\sum_{u\in\tilde{\mathcal{K}}_{s'}} \frac{P_{\max}|h_{u;s}|^2}{2}\right] + \frac{N_0 L}{2} \\ &= \sum_{s'\neq s}\frac{P_{\max}\sigma_h^2\rho N_U}{2N_S} + \frac{N_0 L}{2} \\ &= \frac{P_{\max}\sigma_h^2\rho N_U(N_S-1)}{2N_S} + \frac{N_0 L}{2},\end{aligned} \tag{42}$$

where the inequality is due to (19).

To find the variance of $\mathbf{u}_s$, let

$$\sum_{u\in\tilde{\mathcal{K}}_s}\mathbf{g}_{u;s} = \mathbf{G}_s\mathbf{v}_s,$$

where $\mathbf{G}_s = [\mathbf{g}_{1;s} \ \cdots \ \mathbf{g}_{N_U;s}]$ and $\mathbf{v}_s = [v_{1;s} \ \ldots v_{N_U;s}]^{\mathrm{T}}$. Here, $v_{u;s} = 1$ if $u \in \tilde{\mathcal{K}}_s$ and 0 otherwise. Since the $v_{u;s}$'s are independent with $\Pr(v_{u;s} = 1) = \frac{\rho}{N_S}$, we have

$$\begin{aligned}\mathbb{E}\left[||\mathbf{u}_s||^2\right] &= c^2\mathrm{Tr}\left(\mathbf{G}_s\left(\mathbb{E}\left[\mathbf{v}_s\mathbf{v}_s^{\mathrm{T}}\right] - \mathbb{E}\left[\mathbf{v}_s\right]\mathbb{E}\left[\mathbf{v}_s^{\mathrm{T}}\right]\right)\mathbf{G}_s^{\mathrm{T}}\right) \\ &= c^2\frac{\rho}{N_S}\left(1 - \frac{\rho}{N_S}\right)\mathrm{Tr}\left(\mathbf{G}_s\mathbf{G}_s^{\mathrm{T}}\right) \\ &= c^2\frac{\rho}{N_S}\left(1 - \frac{\rho}{N_S}\right)\sum_{u=1}^{N_S}||\mathbf{g}_{u;s}||^2 \\ &\leq c^2\Delta_m\frac{\rho N_U}{N_S}\left(1 - \frac{\rho}{N_S}\right) \\ &= \kappa\frac{P_{\max}\sigma_h^2\rho N_U}{N_S}\left(1 - \frac{\rho}{N_S}\right).\end{aligned} \tag{43}$$

From (40), (42), and (43), the variance of $\mathbf{w}_s^{(t)}$ as $t \to \infty$ is given by

$$\begin{aligned}\mathrm{Var}(\mathbf{w}_s) &= \mu^2\mathrm{Var}(\mathbf{d}_s) \\ &= \mu^2\left(\mathbb{E}\left[||\mathbf{u}_s||^2\right] + \mathbb{E}\left[||\mathbf{e}_s||^2\right]\right) \\ &\leq \mu^2\frac{P_{\max}\sigma_h^2\rho N_U}{N_S}\left(\frac{N_S - 1}{2} + \kappa\left(1 - \frac{\rho}{N_S}\right)\right) \\ &\quad + \mu^2\frac{N_0 L}{2}.\end{aligned} \tag{44}$$

B. With Randomized Transmission

According to (32), we know that the set of users choosing model m and transmitting their local gradients is given by

$$\begin{aligned}\widehat{\mathcal{K}}_s = \Big\{u :\ &|h_{u;s}|^2 \geq |h_{u;s'}|^2,\ \forall s' \neq s, \\ &\text{and } \frac{c^2}{|h_{u;s}(t)|^2} \leq P_{\max}\Big\},\end{aligned} \tag{45}$$

while $\widehat{\mathcal{K}}_s \subseteq \mathcal{K}_s$. In randomized transmissions, all the transmit powers are the same, i.e., $P_u = P_{\max}$ for all u. However, the average is to be proportional to $g_{u;s}$ due to the on/off transmission. Then, based on (26) and (27), we know that

$$\begin{aligned}\hat{\mathbf{r}}_s = &\sum_{u\in\widehat{\mathcal{K}}_s}\sqrt{P_u}|h_{u;s}|\mathbf{z}_{u;s} \\ &+ \sum_{u'\in\widehat{\mathcal{K}}_s^c}\sqrt{P_{u'}}\sum_{s'\neq s} h_{u';s}\alpha_{u';s'}\mathbf{z}_{u';s'} + \mathbf{n}_s.\end{aligned} \tag{46}$$

Since $h_{u;s}$ and $\mathbf{z}_{u;s}$ are mutually independent and uniformly distributed, $\mathbf{n}_s(t) \sim \mathcal{N}(0, \frac{N_0}{2}\mathbf{I})$, $\hat{\rho} = 1 - \hat{\mathbb{P}}_{\mathrm{out}}$, and $\mathbb{E}[|h_{u;s}|] = \sqrt{\frac{\pi}{2}}\sigma$, according to (27), we have

$$\begin{aligned}\mathbb{E}\left[\hat{\mathbf{r}}_s\right] &= \mathbb{E}\left[\mathbb{E}\left[\hat{\mathbf{r}}_s \,|\, \widehat{\mathcal{K}}_s\right]\right] \\ &= \mathbb{E}\left[\mathbb{E}\left[\sum_{u\in\widehat{\mathcal{K}}_s}\sqrt{P_u}|h_{u;s}|\mathbf{z}_{u;s}\right]\right] \\ &= \mathbb{E}\left[\sqrt{P_{\max}}\mathbb{E}[|h_{u;s}|]\sum_{u\in\widehat{\mathcal{K}}_s}\mathbb{E}\left[\mathbf{z}_{u;s}\right]\right] \\ &= \mathbb{E}\left[\sqrt{P_{\max}}\mathbb{E}\left[|h_{u;s}|\right]\sum_{u\in\widehat{\mathcal{K}}_s}\mathbb{E}\left[\mathbf{z}_{u;s}\right]\right] \\ &= \mathbb{E}\left[\sqrt{\frac{\pi}{2}P_{\max}}\sigma_h\hat{\rho}\sum_{u\in\mathcal{K}_s}\gamma\mathbf{g}_{u;s}\right] \\ &= \frac{\sqrt{\frac{\pi}{2}P_{\max}}\sigma_h\gamma\hat{\rho}N_U}{N_S}\mathbf{g}_s.\end{aligned} \tag{47}$$

Then, as the β_s-smooth assumption, the cost function of the system with randomized transmission, $\widehat{C}_s$, is assumed to be $\hat{\beta}_s$-smooth. We can further assume that $\widehat{C}_s$ is $\hat{\theta}_s$-strongly convex, where $\hat{\theta}_s < \hat{\beta}_s$. Let $\widehat{\mathbf{w}}_s^{(t)} = \mathbb{E}[\mathbf{w}_s^{(t)}]$ and according to the mean

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value theorem, we have

$$||\widehat{\mathbf{w}}_s^{(t+1)} - \mathbf{w}_s^*|| = ||\mathbf{I} - \frac{\sqrt{\frac{\pi}{2}P_{\max}}\sigma_h\gamma\hat{\rho}N_U}{N_S}\nabla^2\widehat{C}_s|| \times ||\widehat{\mathbf{w}}_s^{(t)} - \mathbf{w}_s^*||, \tag{48}$$

where $\nabla^2\widehat{C}_s$ represents the Hessian at $\widehat{\mathbf{w}}_s'$ lies within the line segment that connects between $\widehat{\mathbf{w}}_s^{(t)}$ and $\mathbf{w}_s^*$. Thus, the condition for the convergence in mean can be found as:

$$\max \times \left\{1 - \frac{\sqrt{\frac{\pi}{2}P_{\max}}\sigma_h\gamma\hat{\rho}N_U}{N_S}\hat{\theta}_s, \frac{\sqrt{\frac{\pi}{2}P_{\max}}\sigma_h\gamma\hat{\rho}N_U}{N_S}\hat{\beta}_s - 1\right\} \leq 1. \tag{49}$$

Therefore, with

$$\mu < \hat{\mu} = \frac{2N_S}{\sqrt{\frac{\pi}{2}P_{\max}}\sigma_h\gamma\hat{\rho}N_U(\hat{\theta}_s + \hat{\beta}_s)}, \tag{50}$$

since

$$||\mathbf{I} - \frac{\mu\sqrt{\frac{\pi}{2}P_{\max}}\sigma_h\gamma\hat{\rho}N_U}{N_S}\nabla^2\widehat{C}_s|| < \frac{\hat{\beta}_s - \hat{\theta}_s}{\hat{\beta}_s + \hat{\theta}_s} \leq 1,$$

$\widehat{\mathbf{w}}_s^{(t)} \to \mathbf{w}_s^*$ as $t \to \infty$ from (48) can be confirmed.

The two terms of noise can be expressed as follows:

$$\begin{aligned}\widehat{\mathbf{d}}_s &= \hat{\mathbf{r}}_s - \mathbb{E}[\hat{\mathbf{r}}_s] \\ &= \widehat{\mathbf{u}}_m + \widehat{\mathbf{e}}_s,\end{aligned} \tag{51}$$

where $\widehat{\mathbf{u}}_s = \sqrt{P_{\max}}\sum_{u\in\hat{\mathcal{K}}_s}|h_{u;s}|\mathbf{z}_{u;s} - \mathbb{E}[\hat{\mathbf{r}}_s]$ and

$$\widehat{\mathbf{e}}_s = \sum_{s'\neq s}\sum_{u\in\mathcal{K}_{s'}}\sqrt{P_{\max}}h_{u;s}\alpha_{u;s'}\mathbf{z}_{u;s'} + \mathbf{n}_s. \tag{52}$$

Similar to (42), we have

$$\begin{aligned}\mathbb{E}[||\widehat{\mathbf{e}}_s||^2] &= \mathbb{E}[\mathbb{E}[||\mathbf{e}_s||^2 \mid \widehat{\mathcal{K}}_1, \ldots, \widehat{\mathcal{K}}_{N_S}] \\ &= \sum_{s'\neq s}\mathbb{E}\left[\sum_{u\in\mathcal{K}_{s'}}\frac{P_{\max}||\mathbf{z}_{u;s'}||^2|h_{u;s}|^2}{2}\right] + \frac{N_0L}{2} \\ &\leq \sum_{s'\neq s}\mathbb{E}\left[\sum_{u\in\mathcal{K}_{s'}}\frac{P_{\max}|h_{u;s}|^2}{2}\right] + \frac{N_0L}{2} \\ &= \sum_{s'\neq s}\frac{P_{\max}\sigma_h^2\hat{\rho}N_U}{2N_S} + \frac{N_0L}{2} \\ &= \frac{P_{\max}\sigma_h^2\hat{\rho}N_U(N_S-1)}{2N_S} + \frac{N_0L}{2},\end{aligned} \tag{53}$$

and

$$\begin{aligned}\mathbb{E}[||\widehat{\mathbf{u}}_s||^2] &= P_{\max}\mathbb{E}\left[\left(\sum_{u\in\hat{\mathcal{K}}_s}|h_{u;s}|\mathbf{z}_{u;s} - \frac{\sqrt{\frac{\pi}{2}}\sigma_h\gamma\hat{\rho}N_U}{N_S}\mathbf{g}_s\right)^2\right] \\ &= P_{\max}\left\{\mathbb{E}\left[\left(\sum_{u\in\hat{\mathcal{K}}_s}|h_{u;s}|\mathbf{z}_{u;s}\right)^2\right] + \mathbb{E}\left[\left(\frac{\sqrt{\frac{\pi}{2}}\sigma_h\gamma\hat{\rho}N_U}{N_S}\mathbf{g}_s\right)^2\right] - 2\mathbb{E}\left[\left(\sum_{u\in\hat{\mathcal{K}}_s}|h_{u;s}|\mathbf{z}_{u;s}\right)\frac{\sqrt{\frac{\pi}{2}}\sigma_h\gamma\hat{\rho}N_U}{N_S}\mathbf{g}_s\right]\right\} \\ &= P_{\max}\left\{\mathbb{E}\left[\left(\sum_{u\in\hat{\mathcal{K}}_s}|h_{u;s}|\mathbf{z}_{u;s}\right)\right]^2 + \mathrm{Var}\left(\sum_{u\in\hat{\mathcal{K}}_s}|h_{u;s}|\mathbf{z}_{u;s}\right) - \left(\frac{\sqrt{\frac{\pi}{2}}\sigma_h\gamma\hat{\rho}N_U}{N_S}\mathbf{g}_s\right)^2\right\} \\ &= P_{\max}\mathrm{Var}\left(\sum_{u\in\hat{\mathcal{K}}_s}|h_{u;s}|\mathbf{z}_{u;s}\right) \\ &= P_{\max}\frac{\hat{\rho}}{N_S}\left(1 - \frac{\hat{\rho}}{N_S}\right)\sum_{u=1}^{N_U}||h_{u;s}|\mathbf{z}_{u;s}||^2 \\ &\leq \frac{P_{\max}\sigma_h^2\hat{\rho}N_U}{N_S}\left(1 - \frac{\hat{\rho}}{N_S}\right).\end{aligned} \tag{54}$$

From (51), (53), and (54), the variance of $\widehat{\mathbf{w}}_s^{(t)}$ as $t \to \infty$ can be expressed by

$$\begin{aligned}\mathrm{Var}(\widehat{\mathbf{w}}_s) &= \mu^2\mathrm{Var}(\widehat{\mathbf{d}}_s) \\ &= \mu^2\left(\mathbb{E}\left[||\widehat{\mathbf{u}}_s||^2\right] + \mathbb{E}\left[||\widehat{\mathbf{e}}_s||^2\right]\right) \\ &\leq \mu^2\frac{P_{\max}\sigma_h^2\hat{\rho}N_U}{N_S}\left(\frac{N_S-1}{2} + \left(1 - \frac{\hat{\rho}}{N_S}\right)\right) \\ &\quad + \mu^2\frac{N_0L}{2}.\end{aligned} \tag{55}$$

As discussed above, the variance of the optimal parameter vector of models is the same for all the model, which are influenced by the number of models, N_S, and the number of users, N_U. In addition, with randomized transmission, the variance of the optimal parameter vector is independent of Δ_s. This means that there is an upper bound on the variance of the optimal parameter vector regardless of the transmission method, and the variance of the vector optimization parameters of DOTA-MSRT is not affected by the vector parameters and has stronger stability than AOTA-MS.

VI. Simulation Results

This section presents the simulation results of AOTA-MS and DOTA-MSRT, highlighting their influence on the performance of federated learning. To evaluate the OTA approaches, we focus on the support vector classifier (SVC) using training image data sets in [41]. Specifically, we consider binary linear SVC with only two different classes of image data sets in CIFAR 10. The parameter vector length is determined as $L = 32 \times 32 \times 3 =$

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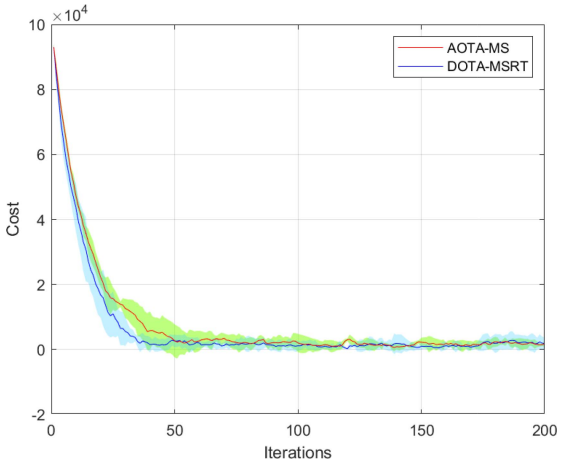


Fig. 2. Learning cost curves with 95% confidence interval (shaded area) when $N_S = 4$, $N_U = 100$, SNR $= 6$ dB, and $\eta = 0.05$.

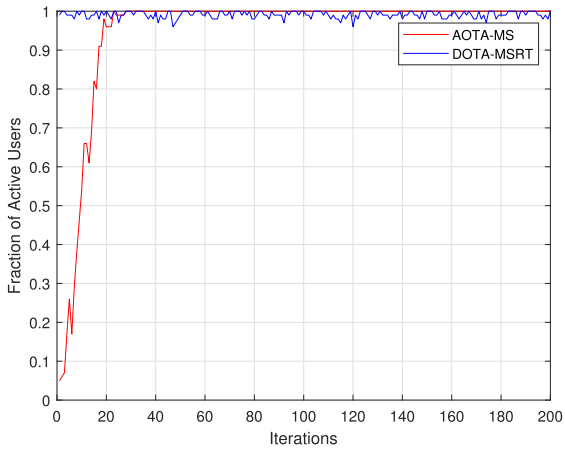


Fig. 3. Fraction of active users with the number of iterations when $N_S = 4$, $N_U = 100$, SNR $= 6$ dB, and $\eta = 0.05$.

3072.[2] As in [42], the cost function based on the hinge function can be shown as

$$C_s(\mathbf{w}, \mathcal{D}_u) = C \max\left(0, 1 - y_u\left(\mathbf{w}^{\mathrm{T}}\mathbf{m}_u - w_0\right)\right) + \lambda||\mathbf{w}||^2, \tag{56}$$

where $y_u \in \{\pm 1\}$ denotes the label of data set at device u (i.e. $y_u = -1$ and $+1$ for labels 0 and 1, respectively), w_0 denotes the offset term, $\lambda = \frac{1}{2} > 0$ is the Lagrange multiplier, and $C = \frac{1}{N_U}$. Thus, the dataset of user u is $\mathcal{D}_u = \{\mathbf{m}_u, y_u\}$. For convenience, the same model in (56) is used for all servers. In addition, suppose that $\sigma_h^2 = 1$ and the SNR is defined as $\mathsf{SNR} = \frac{P_{\max}\sigma_h^2}{N_0} = \frac{P_{\max}}{N_0}$.

Fig. 2 depicts the learning curves, showcasing their convergence performance with a 95% confidence interval (illustrated as a shaded green area) using the cost function in (56). The settings for this plot are $N_S = 4, N_U = 100, \mathsf{SNR} = 6\,\mathrm{dB}$, and $\eta = 0.05$. As observed from Fig. 2, it is evident that DOTA-MSRT achieves a faster convergence speed compared to AOTA-MS. This can be attributed to the fact that DOTA-MSRT has a larger number of active users, i.e., users transmitting local gradient vectors, in the initial stages of training, as compared to AOTA-MS.

Fig. 3 shows the fraction of active users the number of iterations when $N_S = 4$, $N_U = 100$, SNR $= 6$ dB, and $\eta = 0.05$. In the previous discussion, we found that the outage probability of DOTA-MSRT is not affected by the value of the transmission vector, while the outage probability of AOTA-MS is limited by Δ_s. The simulation results depicted in Fig. 3 validate that the number of active users gradually increases and converges in AOTA-MS, whereas it remains constant in DOTA-MSRT. This means that the number of active users will have a certain impact on the convergence speed of federated learning.

Fig. 4 shows the learning cost with the number of APs, N_S, when $N_U = 60$, SNR $= 6$ dB, $\eta = 0.05$, and the number of iterations is equal to 300. This experimental result demonstrates that significantly higher costs are incurred by AOTA - MS than

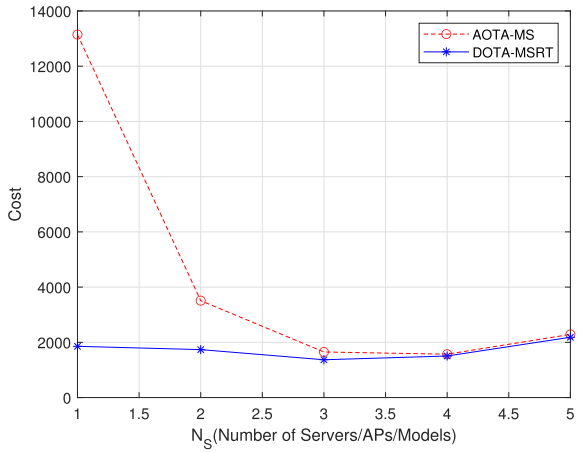


Fig. 4. Learning cost with the number of APs, N_S, when $N_U = 60$, SNR $= 6$ dB, $\eta = 0.05$, and Iterations $= 300$.

DOTA - MSRT when deployed with fewer servers/APs (e.g., $N_S = 1$ or 2), primarily due to its limited model selection diversity during initial training phases. Limited by their own initial value and relatively stable channel state, most users in the AOTA-MS system are difficult to participate in the federated learning process, resulting in limited optimization results and high cost function. Meanwhile, as N_S increases, AOTA-MS demonstrates rapid cost reduction followed by stable convergence. This improvement stems from enhanced collaborative learning opportunities through diversified AP-specific model adaptations. For DOTA-MSRT, the cost function shows gradual decline with increasing N_S when the number of APs is small, as multi-server coordination effectively mitigates training volatility through distributed averaging. In addition, both algorithms exhibit cost function degradation when $N_S \geq 1$. This phenomenon arises from diminished per-AP client density as the system allocates fixed $N_U = 60$ users across more APs, fewer clients participate in each AP's local aggregation per iteration. This reduced participation degrades learning efficiency,

[2] Note that the length of gradient vector is actually $L = 3072 + 1$ due to the offset term.

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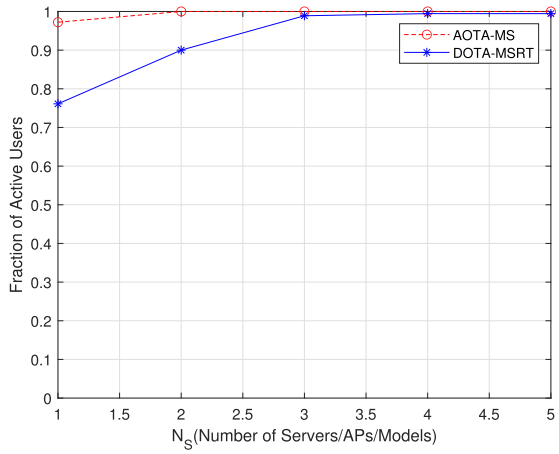


Fig. 5. Fraction of active users with the number of APs, N_S, when $N_U = 60$, $\mathsf{SNR} = 6$ dB, $\eta = 0.05$, and Iterations=300.

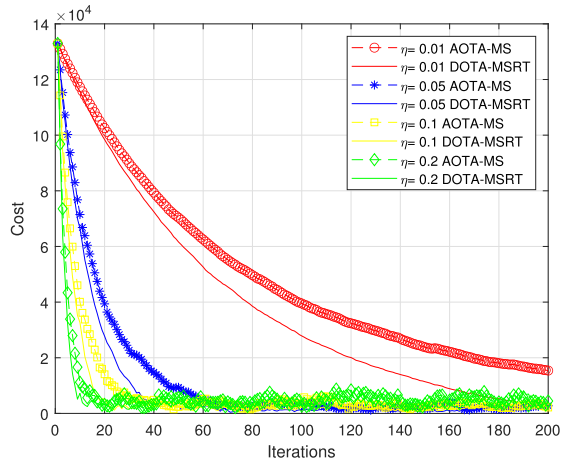


Fig. 7. Learning curves with the length of iterations step, η, when $N_U = 50$, $N_S = 4$, and $\mathsf{SNR} = 6$ dB.

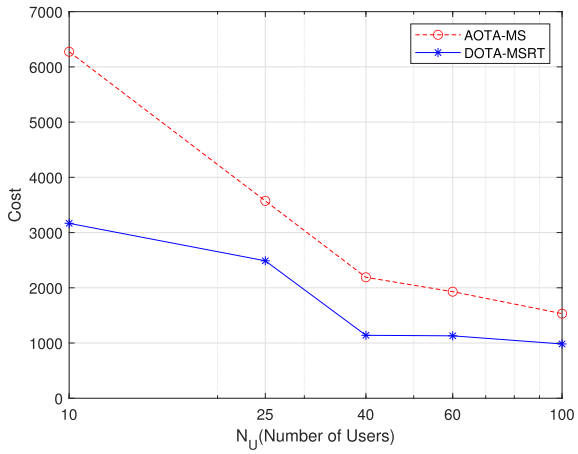


Fig. 6. Learning cost with the number of users, N_U, when $N_S = 5$, $\mathsf{SNR} = 6$ dB, $\eta = 0.05$, and Iterations = 200.

ultimately increasing the cost function despite sufficient computational resources. Crucially, DOTA-MSRT maintains superior cost stability compared to AOTA-MS, attributed to its adaptive mechanism ensuring consistent user participation.

Fig. 5 shows the fraction of active users with the number of APs, N_S, when $N_U = 60$, $\mathsf{SNR} = 6$ dB, $\eta = 0.05$, and the number of iterations is equal to 300. In Fig. 5, we can find that the number of active users increases with N_S in both AOTA-MS and DOTA-MSRT. We can see that as the number of servers N_S increases, the fraction of active users of either AOTA-MS or DOTA-MSRT also increases until the active ratio is 1. As N_S increases, the outage probability of a single user transmitting vector decreases. Meanwhile, the outage probability of DOTA-MSRT is not affected by the value of the transmission vector, while the AOTA-MS outage probability is limited by Δ_s. Therefore, when N_S is low, AOTA-MS may have higher fraction of active users than DOTA after completing iteration. This is consistent with the user outage probability equation, which proves the validity of our analysis.

Fig. 6 shows the learning curves with the number of users, N_U, when $N_S = 5$, $\mathsf{SNR} = 6$ dB, $\eta = 0.05$, and the number of iterations is equal to 200. In Fig. 6, it is evident that AOTA-MS yields higher costs compared to DOTA-MSRT for all N_U, indicating that AOTA-MS performs worse when the number of iterations is not sufficiently large. Moreover, both DOTA-MSRT and AOTA-MS exhibit a decreasing cost trend with increasing N_U due to the increase in the number of active users per model. This suggests that as the number of users N_U grows, each server is more evenly influenced by the user datasets in each iteration.

Fig. 7 shows the learning curves with the length of iteration steps, η, when $N_U = 50$ and $\mathsf{SNR} = 6$ dB. In this figure, it can be observed that when η is larger, the convergence rate of both DOTA-MSRT and AOTA-MS become faster. On the other hand, a smaller value of η keeps the volatility of the learning curve low after convergence. Moreover, for all η, the convergence rate of DOTA-MSRT is better than that of AOTA-MS. Specifically, the number of iterations required for AOTA-MS to converge to stability is about 1.5 times more than that of DOTA-MSRT with the same iteration step. In other words, the convergence speed of DOTA-MSRT is about 1.5 times faster than that of AOTA-MS.

Fig. 8 shows the fraction of active users with the length of iteration steps, η, when $N_U = 50$ and $\mathsf{SNR} = 6$ dB. It demonstrates that a smaller η leads to a slower growth rate of fraction of active users in AOTA-MS, but also to a larger fluctuation in the fraction of active users. It seems that due to the single iteration has less variation for the local user transmitting vector, which leads to a larger randomness in the fraction of active users. However, in DOTA-MSRT, we find that the number of active users basically change a little when the number of iterations increase, indicating that DOTA-MSRT has higher stability in terms of the proportion of active users.

Fig. 9 shows the learning cost with the length of iteration steps, η, when $N_U = 50$, $\mathsf{SNR} = 6$ dB, and the number of iterations is equal to 200. In this figure, for AOTA-MS, the cost is around 8 times higher than that of DOTA-MSRT when $\eta = 0.01$ after 200 iterations, which is caused by the limited convergence rate with low η. When $\eta \geq 0.05$, the cost of AOTA-MS after 200 iterations

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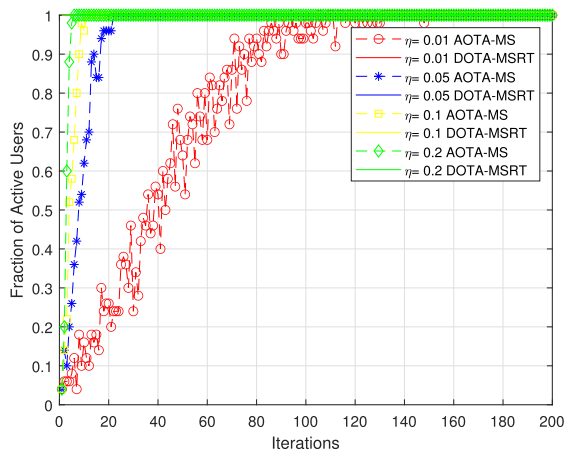


Fig. 8. Fraction of active users as a function of the length of iterations step, η, when $N_U = 50$, $N_S = 4$, and $\mathsf{SNR} = 6$ dB.

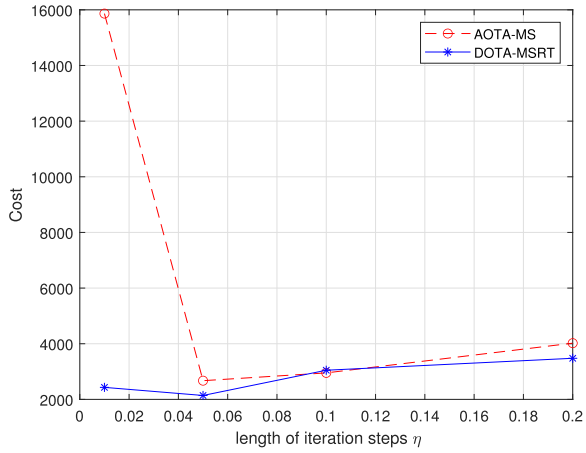


Fig. 9. Learning cost with the length of iterations step, η, when $N_U = 50$, $N_S = 4$, $\mathsf{SNR} = 6$ dB, and Iterations=200.

tends to increase with η. For DOTA-MSRT, although the cost of $\eta = 0.05$ is lower than that of $\eta = 0.01$ due to the convergence rate and the influence of fluctuations, the cost function after 200 iterations still shows an upward trend with the increase of η like AOTA-MS. This is due to the fact that large η tends to bring large fluctuations, resulting in difficulty in approaching the optimal solution. This figure indicates that when the number of iterations is below a certain threshold, a larger η leads to higher convergence rates, but surprisingly does not necessarily yield better convergence results. Meanwhile, it also indicates that AOTA-MS performs worse when the number of iterations is insufficient, which reminds the importance of the convergence rate when aiming to achieve improved communication effectiveness.

VII. Conclusion

In this paper, we have investigated a CP OTA approach for achieving communication-efficient uploading in multi-server federated learning. By leveraging OTA and CP techniques with model selection, we have demonstrated that there is no need for bandwidth expansion when the number of users and models increases. In addition, we have derived key conditions that ensure an unbiased estimate of the aggregation. To overcome the challenges associated with analog OTA, such as the requirement for expensive linear power amplifiers at users, we have introduced a digital OTA approach with randomized transmission. Through simulation results, we have shown that digital OTA offers a lower cost function and requires fewer iterations for convergence compared to analog OTA, primarily due to the ability to accommodate more users simultaneously. Our findings highlight that the proposed DOTA-MSRT better suits the optimal length of iteration steps, effectively balancing the convergence rate and the cost function.

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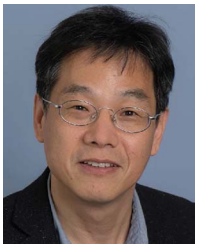


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